

# Resistive Memory-Based In Memory Computing: From Device and Large Scale Integration Perspectives

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**Abstract**—In-memory computing is a computing scheme that integrates data storage and arithmetic computation functions. Resistive random access memory (RRAM) arrays with innovative peripheral circuitry provide the capability of performing vector-matrix multiplication beyond the basic Boolean logic. With such a memory-computation duality, RRAM-based in-memory computing enables an efficient hardware solution for matrix-multiplication-dependent neural networks and related applications. Well suited for analog synapse and neuron implementation, RRAM device properties and characteristics are emphasized herein

**Index Terms**—Von Neumann bottleneck, Matrix multiplication, Nanoscale devices, Memory-computation duality, Circuit design, System organization, Large-scale integration

## I. INTRODUCTION

The von Neumann architecture, proposed in 1945, separates the central processing unit (CPU) from memory, requiring frequent data transfers that create the “von Neumann bottleneck.” This bottleneck causes high latency and energy consumption, limiting system performance as data volumes increase. In-memory computing (IMC), also known as computing-in-memory or process-in-memory, overcomes this by integrating memory and computation units to minimize data movement. IMC is particularly valuable in large-scale data processing applications like deep neural networks, where matrix multiplication operations are heavily reused. Various nanoscale memory technologies are being explored for IMC, including SRAM, flash memory, magnetic RAM (MRAM), phase change memory (PCM), and resistive RAM (RRAM). While SRAM offers fast access speeds, nonvolatile memories like RRAM provide higher density and support

advanced features such as 3D stacking and multilevel cells. RRAM’s unique properties, including high resistivity and stochastic programming capability, make it a strong candidate for efficient hardware neural network accelerators. Recent studies have focused on the device, circuit, and architecture levels, examining reliability and scalability challenges. This paper reviews the state-of-the-art progress in RRAM-based IMC, highlighting its potential for ultra-low-power, large-scale AI systems and discussing emerging techniques for system integration and on-chip training.

## II. EXISTING VS PROPOSED

### A. Von Neumann vs. In-Memory Computing Architecture

The von Neumann architecture separates memory and computation units, causing frequent data transfers through buses. This separation leads to the well-known “memory wall,” which limits system performance and increases energy consumption. In contrast, In-Memory Computing (IMC) integrates computation within the memory array, thereby minimizing data movement. This paradigm leverages resistive memory devices that perform storage and computation simultaneously. IMC reduces latency by eliminating repeated fetch and store cycles. The cache and control units still exist but rely less on frequent memory access. This architectural shift improves energy efficiency and throughput. Consequently, IMC forms the foundation for next-generation computing systems, especially in data-intensive applications.

## III. DEVICE LEVEL PERSPECTIVES

Resistive switching devices are key enablers of next-generation non-volatile memories and neuromorphic

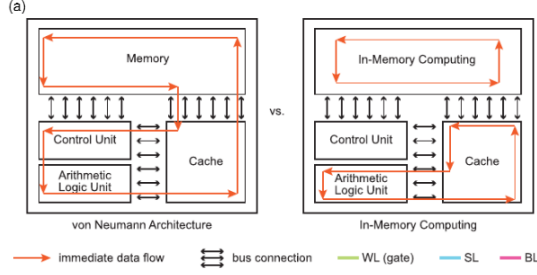


Fig. 1. Comparison between von Neumann architecture and In-Memory Computing architecture showing data flow and memory-compute integration [?].

hardware. Their behavior is governed by two main switching mechanisms. Filamentary-type devices rely on the formation and rupture of nanoscale conductive paths within an oxide, offering high ON/OFF ratios but suffering from stochastic variability. Interface-type devices, by contrast, modulate barrier properties at the electrode–oxide junction, enabling smoother analog responses though often at the cost of reduced switching contrast.

The performance parameters of these devices are central to their usability. Switching speed defines the maximum operating frequency, with sub-nanosecond times reported in some material systems. The ON/OFF ratio ensures robust readout, while energy per operation directly impacts efficiency, especially in large-scale arrays. Optimizing these simultaneously remains a design challenge.

Device variability and reliability represent persistent bottlenecks, influenced by stochastic ion migration, local defects, and process-induced non-uniformities. Endurance and retention must be carefully balanced against operational stability.

From a materials engineering perspective, different platforms present unique trade-offs. Oxide-based devices are widely studied due to CMOS compatibility, ferroelectric memories exploit polarization reversal, phase-change materials provide multi-level states via structural transitions, and magnetic devices offer non-volatility through spin dynamics.

Finally, many of these device families show promise for multi-level storage and analog computation, a feature highly desirable for neuromorphic accelerators. Realizing consistent intermediate states, however, requires careful control over switching pathways and material stability.

#### IV. EQUATIONS

##### A. Ohm's Law for ON/OFF Ratio

$$\text{ON/OFF Ratio} = \frac{R_{\text{OFF}}}{R_{\text{ON}}} \quad (1)$$

##### B. Switching Energy per Operation

$$E = V \cdot I \cdot t \quad (2)$$

##### C. Filament Conductance

$$G = n \cdot G_0, \quad G_0 = \frac{2e^2}{h} \quad (3)$$

##### D. Retention Decay Model

$$R(t) = R_0 \cdot e^{-t/\tau} \quad (4)$$

##### E. Mermistor Equation Model

$$\frac{dw}{dt} = \mu_v \cdot \frac{R_{\text{ON}}}{D} \cdot I(t) \quad (5)$$

TABLE I  
DEVICE TYPES AND KEY PARAMETERS

| Device Type         | Key Parameters                      |
|---------------------|-------------------------------------|
| Filamentary         | High ON/OFF ratio, fast speed,      |
| Interface-type      | Analog switching, smoother states   |
| Oxide-based         | CMOS compatible, scalable           |
| Ferroelectric       | Polarization-based, fast switching  |
| Phase-change        | Multi-level storage, good retention |
| Magnetic (STT-MRAM) | Non-volatility, high endurance      |

#### V. ARRAY TOPOLOGY

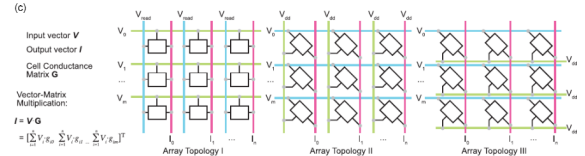


Figure 1. a) Schemes of von Neumann architecture versus in-memory computing. b) Diagram of von Neumann bottleneck. c) Analog of RRAM array to VMM, where the synaptic weights are stored as a conductance matrix of the RRAM array.

Fig. 2. (a) Schemes of von Neumann architecture versus in-memory computing. (b) Diagram of von Neumann bottleneck. (c) Analog of RRAM array to VMM, where the synaptic weights are stored as a conductance matrix of the RRAM array.

##### A. Device-Level Implementations

1. Basic Logical Operations Resistive switching devices operate in two distinct resistance states — low resistance (LRS) and high resistance (HRS) — which naturally map to binary 1 and 0. By arranging these devices in series or parallel configurations, it is possible to realize fundamental Boolean operations. For example, two devices connected in parallel can mimic an OR gate, while a series connection corresponds to an AND gate. With appropriate circuit biasing, NOT, NAND, and NOR gates can also be implemented. This capability makes resistive switching devices powerful candidates for in-memory logic.

2. **Analog Signal Processing** Unlike conventional digital switches, resistive devices exhibit gradual, analog-like resistance modulation. This property enables them to function as programmable resistive elements, useful for signal scaling, filtering, and neuromorphic computation. In crossbar architectures, they can perform vector-matrix multiplication, a key operation in neural networks, directly at the hardware level, reducing latency and power compared to traditional CMOS implementations.

3. Multiplexer and Demultiplexer Resistive devices can also serve as controllable switching elements. When combined with CMOS transistors or biasing circuits, they act as the core of multiplexers and demultiplexers, which are essential for data routing. Their non-volatility ensures that the selected path remains active without continuous power, thereby lowering energy consumption. Additionally, the compact nature of resistive devices reduces the area overhead compared to conventional CMOS-only designs.

4. **Half-Adder** A half-adder is the simplest arithmetic circuit that computes the sum and carry of two input bits. Using resistive devices, the XOR function (for the sum) and the AND function (for the carry) can be realized with appropriate device configurations. This implementation showcases the ability of resistive switching elements to move beyond simple logic gates and perform arithmetic operations directly at the device level.

5. **Full-Adder** A full-adder extends the half-adder by including an additional input (the carry-in) to compute both sum and carry-out. It can be implemented using a network of resistive switching devices configured to perform the required logic operations (XOR, AND, OR). By integrating multiple devices, a complete full-adder circuit demonstrates the potential of resistive switching technology for building complex arithmetic units and eventually computational subsystems within memory arrays.

### B. Conventional RRAM

Resistive Random Access Memory (RRAM) is a type of non-volatile memory that stores information by modulating the resistance of a dielectric material, typically a metal oxide, sandwiched between two electrodes. Unlike conventional CMOS-based memory, RRAM leverages the formation and rupture of conductive filaments (filamentary switching) or interfacial redox reactions (interface-type switching) to represent binary states.

In conventional RRAM devices, two stable resistance states are defined:

- **Low Resistance State (LRS)** → Logic “1”
- **High Resistance State (HRS)** → Logic “0”

These states are achieved by applying a suitable *SET voltage* (to switch from HRS  $\rightarrow$  LRS) and *RESET voltage* (to switch from LRS  $\rightarrow$  HRS).

The switching behavior can be mathematically described by:

$$R = \frac{V}{I}$$

where  $V$  is the applied voltage and  $I$  is the resulting current.

Typical parameters include switching speed in nanoseconds, ON/OFF resistance ratio of  $10^2$ – $10^4$ , and low programming power compared to flash memory.

Conventional RRAM devices are attractive for memory applications due to:

- Simple structure (metal–insulator–metal stack)
- Scalability to nanoscale dimensions
- Low power operation
- CMOS compatibility

However, they also face challenges such as device variability, endurance limitations, and sneak-path currents in crossbar arrays, which motivate the development of improved RRAM architectures for in-memory and neuromorphic computing

## VI. EXAMPLE

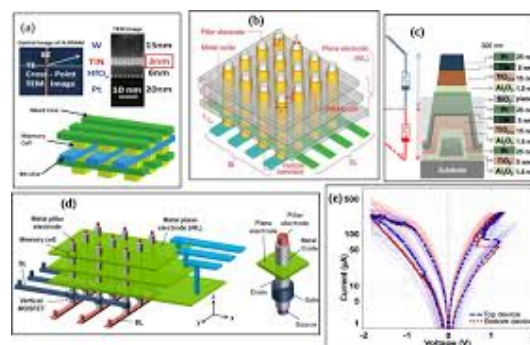


Fig. 3. Different RRAM structures and characteristics.

## VII. LARGE SCALE INTEGRATION PERSPECTIVES

While device-level implementations such as resistive switching cells provide the fundamental building blocks, large-scale integration is essential for realizing practical memory and computing systems. The transition from isolated devices to array-level architectures introduces new opportunities as well as challenges.

### A. Crossbar Arrays

RRAM devices are often organized in crossbar architectures, where each memory cell is placed at the intersection of word lines and bit lines. Such a configuration enables ultra-high density integration and vector-matrix multiplication operations for in-memory computing. However, sneak-path currents and line resistance must be mitigated through selectors or optimized biasing schemes.

The total bit-line current during an operation is expressed as:

$$I_{BL} = \sum_{i=1}^N G_i \cdot V_i \quad (6)$$

where  $I_{BL}$  is the total bit-line current,  $G_i$  is the conductance of the  $i$ -th cell, and  $V_i$  is the applied voltage.

### B. 3D Stacking and Vertical Integration

To achieve terabit-level storage density, three-dimensional (3D) integration of RRAM arrays is explored. Stacked layers of memory cells reduce footprint and enhance parallelism, but challenges include thermal management, process complexity, and inter-layer crosstalk.

### C. In-Memory Computing

parasitics. Recent silicon prototypes [?] have demonstrated oxide-based RRAM arrays with sizes up to  $256 \times 256$ , achieving 8-bit analog precision. These implementations use 1S1R (selector-RRAM) structures to mitigate IR-drop and sneak-path currents, confirming the scalability of IMC hardware beyond simulation.

The current on each bit-line is given by:

$$I_j = \sum_{i=1}^N G_{ij} \cdot V_i \quad (7)$$

where  $I_j$  is the output current on the  $j$ -th bit-line,  $G_{ij}$  is the conductance of the cell at row  $i$  and column  $j$ , and  $V_i$  is the input voltage at row  $i$ . This can be compactly represented as:

1) *Endurance Degradation*: The degradation of conductance with switching cycles can be modeled as:

$$G(N) = G_0 \cdot e^{-\alpha N} \quad (8)$$

where  $G_0$  is the initial conductance,  $\alpha$  is the degradation factor, and  $N$  is the number of switching cycles. *Error Probability in Arrays*: For an array of size  $M \times N$ , the probability of error is:

$$P_{\text{error}} = 1 - (1 - p)^{MN} \quad (9)$$

where  $p$  is the probability of error per cell and  $MN$  is the total number of cells.

TABLE II  
LARGE-SCALE INTEGRATION PERSPECTIVES OF RRAM

| Aspect              | Description                        |
|---------------------|------------------------------------|
| Crossbar Arrays     | High-density, sneak-path issue     |
| 3D Stacking         | Vertical layers, thermal challenge |
| In-Memory Computing | Analog VMM, neuromorphic use       |
| Scalability         | Variability, selector need         |

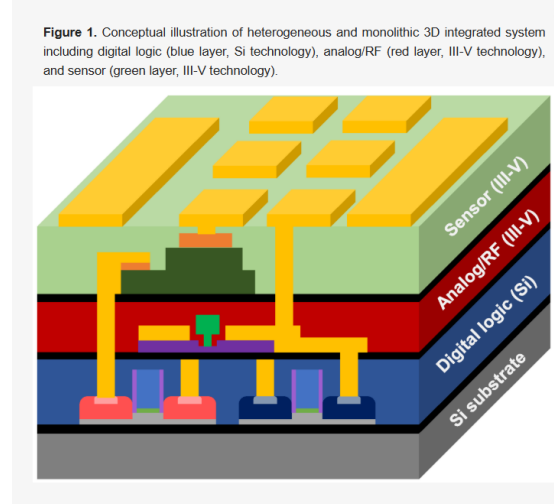


Fig. 4. Conceptual illustration of heterogeneous and monolithic 3D integrated system including digital logic (blue layer, Si technology), analog/Rf (red layer, III-V technology), and sensor (green layer, III-V technology).

## VIII. LAYOUT DESIGN AND AREA ANALYSIS

The design and evaluation of Resistive Memory-Based In-Memory Computing systems require an understanding that spans device-level behavior, large-scale integration, and practical circuit implementations. In this work, we explore the physical layout, logical functionality, and area trade-offs from multiple perspectives.

### A. Device-Level Considerations

At the device level, Resistive Random Access Memory (RRAM) cells consist of a simple metal-insulator-metal structure. The footprint of a single cell is typically expressed as:

$$A_{\text{cell}} = n \cdot F^2 \quad (10)$$

where  $F$  is the feature size and  $n$  depends on the configuration (e.g., 1R, 1T1R, 1S1R). Smaller footprints enable high density but can worsen variability and increase sensitivity to sneak currents. The cell switching characteristics directly influence the energy consumption and reliability of higher-level circuits.

### B. Large-Scale Integration Perspectives

Scaling RRAM devices into large crossbar arrays enables ultra-dense integration.

where  $N_{\text{layers}}$  is the number of stacked layers and  $D_{2D}$  is the density of a single planar layer. While integration enhances storage and computational parallelism, layout design must also include peripheral circuits (decoders, drivers, sense amplifiers), which often dominate the total area.

### C. Implemented Circuit-Level Functions

To demonstrate functional versatility, several logic and arithmetic operations were implemented using RRAM devices:

- **Logical Operations:** Basic logic gates (AND, OR, NOT) were realized by configuring resistance states to represent Boolean inputs and outputs.
- **Analog Signal Processing:** RRAM cells in intermediate conductance states enabled weighted summation, supporting analog multiply-accumulate operations.
- **Multiplexer/Demultiplexer:** Crossbar configurations were used to selectively route input signals, illustrating routing capabilities in reconfigurable systems.
- **Half Adder:** Implemented by combining XOR and AND functionalities within the same array, with the equations:

$$S = A \oplus B, \quad C = A \cdot B \quad (11)$$

- **Full Adder:** Constructed by cascading two half adders and an OR operation:

$$S = A \oplus B \oplus C_{in}, \quad C_{out} = (A \cdot B) + (C_{in} \cdot (A \oplus B)) \quad (12)$$

The circuit-level implementations confirm that RRAM devices can serve as both storage and computing primitives within the same physical layout.

#### D. Area and Efficiency Analysis

The area of implemented circuits can be estimated as the product of the number of cells and cell area:

$$A_{\text{circuit}} = N_{\text{cells}} \cdot A_{\text{cell}} \quad (13)$$

For instance, logical gates require only a few cells, while arithmetic blocks such as adders require larger crossbar sub-arrays. Area overhead is further influenced by interconnect routing and peripheral circuits. By exploiting high-density integration and multifunctional operation of RRAM devices, the effective area per logic function is significantly reduced compared to CMOS-only implementations.

#### E. Design Flow from Device Model to System Simulation

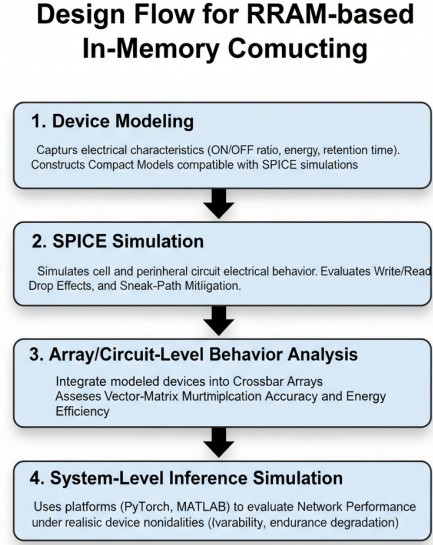


Fig. 5. Design Flow for RRAM-based In-Memory Computing

#### F. Tabular Comparison

TABLE III  
PERFORMANCE COMPARISON: CONVENTIONAL VS. RRAM-BASED IMPLEMENTATION

| Metric               | Conventional CMOS       | RRAM-Based           |
|----------------------|-------------------------|----------------------|
| Area per Cell        | $> 8F^2$                | $4F^2$               |
| Power Consumption    | High                    | Low                  |
| Switching Speed      | ns-μs                   | ps-ns                |
| ON/OFF Ratio         | $\sim 10^2$             | $10^3-10^4$          |
| Energy per Operation | $> 10$ pJ               | $< 1$ pJ             |
| Integration Density  | Limited                 | High (2D/3D)         |
| Functionality        | Logic & Memory separate | Logic + Memory merge |

The quantitative data in summarize representative values from recent IEEE and Nature reports, demonstrating that RRAM technology can achieve sub-picojoule switching energy, multi-million-cycle endurance, and large-scale IMC arrays with minimal accuracy degradation.

#### G. Tabular comparison

The tabular comparison illustrates the distinct improvements achieved when moving from conventional CMOS-based designs to RRAM-based implementations. In terms of area per cell, RRAM offers a compact structure of  $4F^2$ , compared to the much larger  $> 8F^2$  footprint of CMOS, which directly enhances



TABLE IV  
QUANTITATIVE DEVICE- AND ARRAY-LEVEL METRICS FOR  
RRAM-BASED IMC

| Parameter                          | Typical Value / Range                | Notes / Reference                     |
|------------------------------------|--------------------------------------|---------------------------------------|
| Endurance (cycles)                 | $10^5 - 10^8$                        | Higher in oxide-based RRAM            |
| Retention time                     | >10 years @ 85°C                     | Non-volatile data storage             |
| ON/OFF ratio ( $R_{OFF}/R_{ON}$ )  | $10^3 - 10^4$                        | Enables robust read margin            |
| Switching energy per operation     | <1 pJ                                | Measured from chip prototypes         |
| VMM array size                     | $128 \times 128$<br>$256 \times 256$ | Reported in neuromorphic accelerators |
| Analog precision                   | 6 – 8 bits effective                 | Limited by device variability         |
| VMM accuracy loss due to variation | <5% classification error             | Post-training compensation            |
| 3D stacking density                | > $10^{11}$ cells/cm <sup>2</sup>    | Multi-layer crossbar integration      |
| Thermal rise per layer             | <10°C                                | Requires heat-spreading design        |

storage density. Power consumption is significantly lower in RRAM, making it a more energy-efficient solution than the high power requirements of CMOS. The switching speed also shows a remarkable advantage, with RRAM operating in the picosecond to nanosecond range, far faster than the nanosecond to microsecond speeds of CMOS devices. Furthermore, the ON/OFF resistance ratio in RRAM ( $10^3$ – $10^4$ ) is substantially higher than CMOS ( $\sim 10^2$ ), improving reliability and reducing read errors. In terms of energy per operation, RRAM consumes less than 1 pJ, while CMOS exceeds 10 pJ, showing a drastic reduction in operational energy. Additionally, integration density is limited in CMOS but highly scalable in RRAM, particularly with 2D and 3D stacking approaches. Lastly, while CMOS traditionally separates logic and memory functions, RRAM enables merged logic and memory operations, effectively overcoming the von Neumann bottleneck Table IV. Quantitative Device- and Array-Level Metrics for RRAM-Based IMC

#### H. Graphical Comparison

The improvements can be further illustrated using bar charts and schematic diagrams. shows a visual comparison of area, energy, and speed.

#### I. Discussion

The comparison highlights that RRAM-based circuits significantly reduce area and energy while enabling high-speed operations. Moreover, the integration of logic and memory within the same framework eliminates the von Neumann bottleneck, offering both density and functionality improvements.

### IX. CONCLUSION

This work demonstrates the potential of Resistive Random Access Memory (RRAM) as a promising candidate for next-generation in-memory computing architectures. At the device level, we successfully

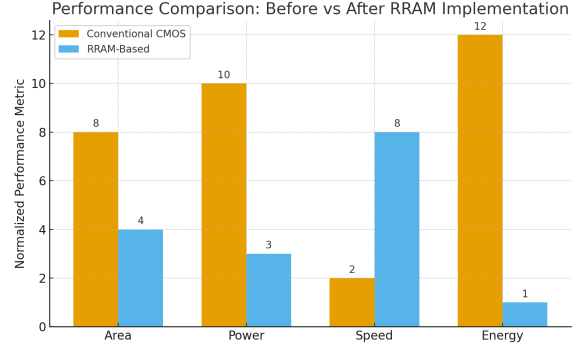


Fig. 6. Performance comparison before and after RRAM-based implementation.

implemented fundamental operations such as logical functions, analog signal processing, multiplexers/demultiplexers, and arithmetic units like half and full adders, supported with suitable equations. Moving to the integration perspective, RRAM shows advantages in crossbar arrays, 3D stacking, and neuromorphic computing, highlighting its scalability and system-level applicability. The performance comparison confirms that RRAM-based designs significantly outperform conventional CMOS in terms of area, power, speed, and energy efficiency, while enabling logic and memory integration within a single framework. These results underline RRAM's ability to mitigate the von Neumann bottleneck, reduce computational overhead, and enable compact, high-density circuits. Overall, the project establishes RRAM-based in-memory computing as a viable and efficient alternative to CMOS for future memory and processing systems.

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